Powers of i and the Complex Plane
Answer Key

Algebra 2

Quick Answers

1. i squared = -1, i cubed = $-i$, i to the fourth = 1
2. $-i$
3. -z = $-2-5i$, 2z = $4+10i$, —
4. 0
5. iz = $-1+3i$, —
6. $-3+5i$
7. w in standard form = $-4+3i$, distance from the origin = 5, —
8. the new position = $-6+2i$, —
9. $-1+32i$
10. the final position = $2-5i$, —

What is verified

14 of 19 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problems 3, 5, 7, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2
HSN-CN.A.1 – *problems 1, 2, 4, 6, 9*
HSN-CN.B.4 – *problems 3, 5, 7, 8, 10*
Difficulty 1–5 of 5 across 19 tagged checks.

1. Complete the cycle table for i^1 through i^4.

Solution: Each entry is the previous one multiplied by i. The definition of the imaginary unit gives the second entry directly: $\boxed{i^2 = -1}$

Multiplying again, $i^2 \cdot i = -1 \cdot i$, so $\boxed{i^3 = -i}$

Once more, $i^3 \cdot i = -i \cdot i = -i^2$, so $\boxed{i^4 = 1}$

After the fourth power the values repeat in the same order, which is the fact every later problem on this sheet uses.

2. Simplify i^{27}.

Solution: Divide the exponent by 4: $27 = 4 \cdot 6 + 3$, so $i^{27} = (i^4)^6 \cdot i^3 = 1 \cdot i^3$. From the table, $i^3 = -i$. $\boxed{-i}$

3. $z = 2 + 5i$. (a) Write $-z$ and $2z$ in standard form. (b) Plot z, $-z$ and $2z$.

Solution (a): Negating a complex number negates both parts, so $\boxed{-z = -2 - 5i}$

Multiplying by the real number 2 scales both parts, so $\boxed{2z = 4 + 10i}$

Part (b) — instructor-judged. On the plane, z is the point $(2, 5)$, $-z$ is $(-2, -5)$ — the reflection of z through the origin — and $2z$ is $(4, 10)$, on the same ray from the origin but twice as far out. All three points must be plotted and labelled. Grade the plot against the values the student wrote in part (a).

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4. Simplify $i^{40} + i^{22}$.

Solution: 40 is a multiple of 4, so $i^{40} = 1$. For 22, $22 = 4 \cdot 5 + 2$, so $i^{22} = i^2 = -1$. Adding, $1 + (-1)$. $\boxed{0}$

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5. $z = 3 + i$. (a) Compute iz. (b) Plot z and iz and describe what multiplying by i did.

Solution (a): Distribute and replace i^2 with -1: $i(3 + i) = 3i + i^2 = -1 + 3i$.

$\boxed{iz = -1 + 3i}$

Part (b) — instructor-judged. z is the point $(3, 1)$ and iz is $(-1, 3)$. A correct description says that multiplying by i turned the point a quarter turn (90°) counterclockwise about the origin, and that its distance from the origin is unchanged — both points are $\sqrt{10}$ from the origin. Describing only the sign changes, without naming a turn about the origin, does not earn the mark.

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6. Simplify $3i^{14} - 5i^7$.

Solution: Reduce each exponent modulo 4.

$$i^{14} = i^{12} \cdot i^2 = i^2 = -1$$
$$i^7 = i^4 \cdot i^3 = i^3 = -i$$

So the expression is $3(-1) - 5(-i)$. $\boxed{-3 + 5i}$

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7. w sits 4 units left of the origin and 3 units above it. (a) Write w in standard form and give its distance from the origin. (b) Plot w.

Solution (a): Left is the negative real direction and up is the positive imaginary direction, so the point is $(-4, 3)$ and $\boxed{w = -4 + 3i}$

The distance from the origin is the hypotenuse of a right triangle with legs 4 and 3, namely $\sqrt{4^2 + 3^2} = \sqrt{25}$, so the distance is $\boxed{5}$

Part (b) — instructor-judged. The point must be marked four units left and three units up, with the segment to the origin drawn. Credit the plot against the number the student wrote in the first blank.

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8. A point at $z = 6 - 2i$ is turned a quarter turn counterclockwise twice, i.e. multiplied by i^2. Give the new position and explain why two quarter turns negate both parts.

Solution: $i^2 = -1$, so the new position is $-(6-2i)$. $\boxed{-6+2i}$

Written part — instructor-judged. A full-credit explanation says that one multiplication by i is a quarter turn about the origin, so two of them make a half turn of $180°$; a half turn sends every point to the opposite side of the origin, which is exactly what negating both parts does. Stating $i^2 = -1$ without connecting it to the turn does not earn the mark.

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9. Simplify $(2i)^5 + i^{50}$.

Solution: Raise both factors of $2i$ to the fifth power first.

$$(2i)^5 = 2^5 \cdot i^5 = 32 \cdot i^4 \cdot i = 32i$$
$$i^{50} = i^{48} \cdot i^2 = -1$$

Adding the two results gives the standard form. $\boxed{-1+32i}$

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10. A robot arm at $z = 5+2i$ is turned a quarter turn counterclockwise three times. (a) Final position? (b) Plot start and finish and justify the use of i^3.

Solution (a): Three quarter turns is multiplication by i three times, that is by $i^3 = -i$. Distributing, $-i(5+2i) = -5i - 2i^2$, and $-2i^2 = +2$. $\boxed{2-5i}$

Part (b) — instructor-judged. The start is $(5,2)$ and the finish is $(2,-5)$; both must be plotted. A full-credit justification says that each quarter turn is one multiplication by i, so three turns in a row multiply by i three times, which is i^3. A student who adds that three quarter turns counterclockwise land in the same place as one quarter turn clockwise has seen the extra structure. Arithmetic with no statement linking repeated turning to the repeated factor does not earn the justification mark.